

Dialogue Systems

Approaches for Conversational Chatbots

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آکادمی‌ها



➤ Rule-base Models

- ▶ ELIZA

- ▶ PARRY

➤ Corpus-based

- ▶ Retrieval Methods

- ▶ Generation Methods



ELIZA: Weizenbaum (1966)

...

You are like my father in some ways.

WHAT RESEMBLANCE DO YOU SEE

You are not very aggressive but I think you don't want me to notice that.

WHAT MAKES YOU THINK I AM NOT AGGRESSIVE

You don't argue with me.

WHY DO YOU THINK I DON'T ARGUE WITH YOU

You are afraid of me.

DOES IT PLEASE YOU TO BELIEVE I'M AFRAID OF YOU

My father is afraid of everybody.

WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER

Bullies.

DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR BOYFRIEND

MADE YOU COME HERE



- Be a Rogerian psychologist
- Draw the patient out by reflecting patient's statements back at them
- Rare type of conversation in which one can "assume the pose of knowing almost nothing of the real world"



Rogerian Psychologist

Patient: "I went for a long boat ride"

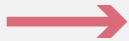
Psychologist: "Tell me about boats"

- You don't assume she didn't know what a boat is
- You assume she had some conversational goal
- Chatbots trying to pass the Turing test usually choose some such domain



Eliza Pattern/Transform Rules

(0 YOU 0 ME) [pattern]



(WHAT MAKES YOU THINK I 3 YOU) [transform]

0 means Kleene *

The 3 is the constituent # in pattern

You hate me

WHAT MAKES YOU THINK I HATE YOU



- Rules are organized by **keywords**
- Each keyword has a pattern and a list of possible transform

Keyword: YOU

Pattern: (O YOU O ME)

Transforms: (WHAT MAKES YOU THINK I 3 YOU)

 (WHY DO YOU THINK I 3 YOU)

keyword pattern list of transforms

$$\begin{pmatrix} K & ((D_1) & (R_{1,1}) & (R_{1,2}) & \cdots & (R_{1,m_1})) \\ & ((D_2) & (R_{2,1}) & (R_{2,2}) & \cdots & (R_{2,m_2})) \\ & \vdots & & & & \vdots \\ & ((D_n) & (R_{n,1}) & (R_{n,2}) & \cdots & (R_{n,m_n})) \end{pmatrix})$$



➤ "I" is very general:

ELIZA: YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

➤ "Everybody" is more specific and interesting

ELIZA: WHO IN PARTICULAR ARE YOU THINKING OF?

➤ **Implementation: keywords stored with their rank**

0 (list of transformation rules)



ELIZA Generator

function ELIZA GENERATOR(*user sentence*) **returns** *response*

Find the word w in *sentence* that has the highest keyword rank

if w exists

Choose the highest ranked rule r for w that matches *sentence*

$response \leftarrow$ Apply the transform in r to *sentence*

if $w = \text{'my'}$

$future \leftarrow$ Apply a transformation from the 'memory' rule list to *sentence*

Push *future* onto memory stack

else (no keyword applies)

either

$response \leftarrow$ Apply the transform for the NONE keyword to *sentence*

or

$response \leftarrow$ Pop the top response from the memory stack

return(*response*)



PLEASE GO ON

THAT'S VERY INTERESTING

I SEE



(MEMORY MY

(0 MY 0 = LETS DISCUSS FURTHER WHY YOUR 3)

(0 MY 0 = EARLIER YOU SAID YOUR 3)

(0 MY 0 = DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR 3))

➤ Whenever "MY" is highest keyword

- ▶ Randomly select a transform on the MEMORY list
- ▶ Apply to sentence
- ▶ Store on a (first-in-first-out) queue

➤ Later, if no keyword matches a sentence

- ▶ Return the top of the MEMORY queue instead



Ethical Implications: Anthropomorphism and Privacy

- People became deeply emotionally involved with the program
- One of Weizenbaum's staff asked him to leave the room when she talked with ELIZA
- When he suggested that he might want to store all the ELIZA conversations for later analysis, people immediately pointed out the privacy implications
 - ▶ Suggesting that they were having quite private conversations with ELIZA
 - ▶ Despite knowing that it was just software.



➤ **Rule-base Models**

▶ ELIZA

▶ PARRY

➤ **Corpus-based**

▶ Retrieval Methods

▶ Generation Methods



PARRY: A Computational Model of Schizophrenia

➤ Another chatbot with a clinical psychology focus

Colby, K. M., Weber, S., and Hilf, F. D. (1971). Artificial paranoia. *Artificial Intelligence* 2(1), 1–25.

➤ Used to study schizophrenia

➤ Same pattern-response structure as Eliza

➤ But a much richer:

- ▶ Control structure

- ▶ Language understanding capabilities

- ▶ Model of mental state.

- ▶ Variables modeling levels of Anger, Fear, Mistrust



Affect Variables

Fear (0-20)

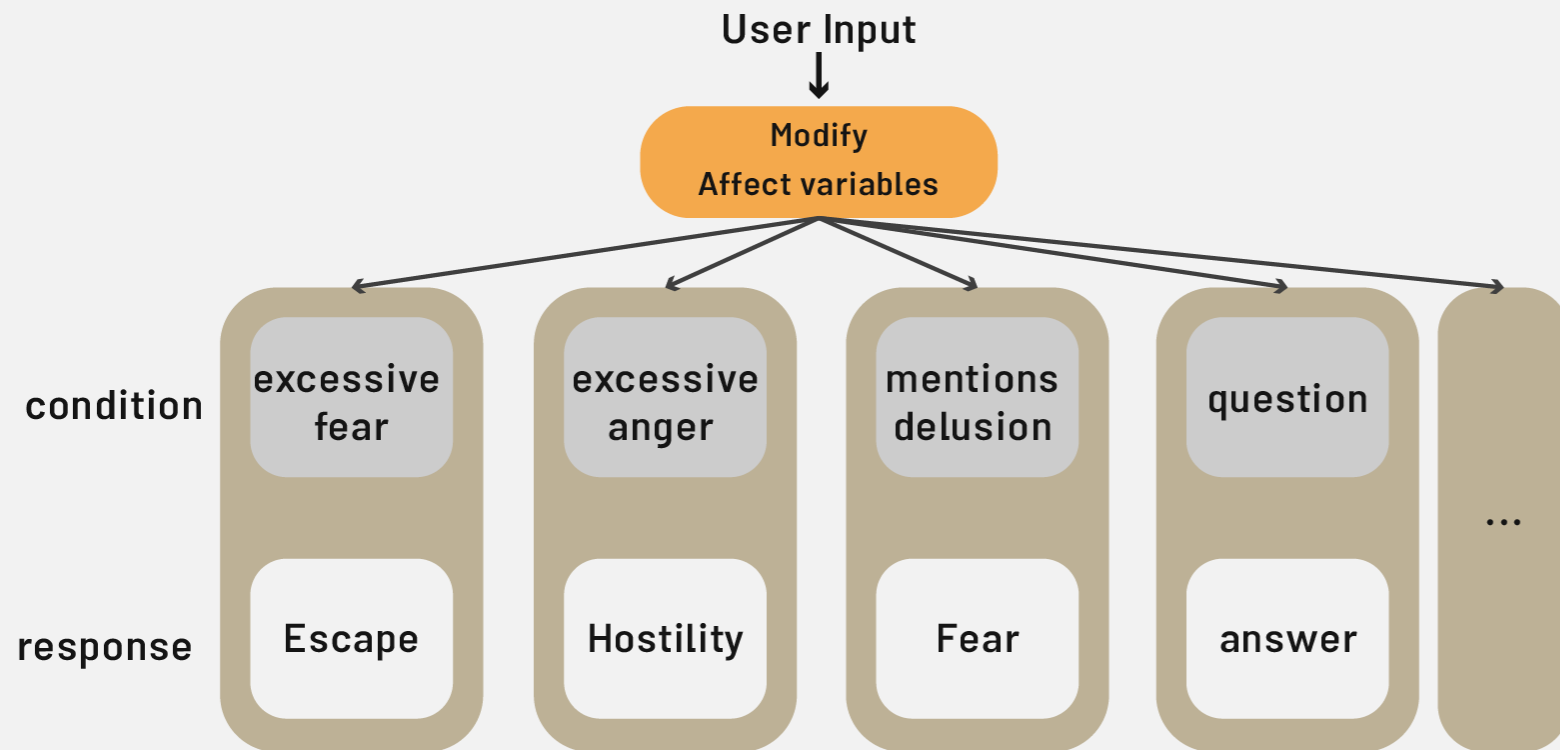
Anger (0-20)

Mistrust (0-15)

- Start with all variables low
- After each user turn
 - ▶ Each user statement can change Fear and Anger
 - ▶ E.g., Insults increases Anger, Flattery decreases Anger
 - ▶ Mentions of his delusions increase Fear
 - ▶ Else if nothing malevolent in input
 - ▶ Anger, Fear, Mistrust all drop



PARRY's Responses Depend on Mental State





PARRY Passes the Turing Test in 1972

- The first system to pass a version of the Turing test
- Psychiatrists couldn't distinguish interviews with PARRY from (text transcripts of) interviews with people diagnosed with paranoid schizophrenia

Colby, K. M., Hilf, F. D., Weber, S., and Kraemer, H. C. (1972). Turing-like indistinguishability tests for the validation of a computer simulation of paranoid processes.

Artificial Intelligence 3, 199–221.



Ethical Implications

- It worried Weizenbaum that people confided in ELIZA
 - ▶ Were people misled about how much computers understood?
- Turkle studied users of ELIZA and other systems
 - ▶ Turkle has shown human face-to-face interaction is vital
 - ▶ But people also develop specific relationships with artifacts
 - ▶ Some users told her ELIZA was more like a kind of diary, a way to privately explore their thoughts.
- Importance of **value-sensitive design**

Joseph Weizenbaum. 1976. Computer Power and Human Reason: From Judgment to Calculation. WH Freeman.

Sherry Turkle. 2011. Taking Things at Interface Value, chapter in Life on the Screen. Simon and Schuster.

Sherry Turkle. 2007. Authenticity in the age of digital companions. Interaction Studies, 8(3), pp.501-517